



Salvage C175 Engine Cylinder Block Upper and Lower Liner Bores Using Metal Spray

Maintenance and Repair	Application	Component Life Management	Component Renewal (CRC)	MARC Management
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1.0 Introduction

Significant cost savings can be achieved by salvaging the upper- and lower-cylinder block liner bores of the C175 engine cylinder block using metal spray technology.



Figure 1. Pitting on upper- and lower-cylinder block liner bores

2.0 Best Practice Description

This process has been developed as a superior salvage procedure. This procedure eliminates the need to install a machined sleeve when pitting exists on the upper- and lower-cylinder block liner bores in C175 engines.

Upper- and lower-cylinder block liner bores salvage procedure consists of: Machining the cylinder block liner bores to remove the pitting, grit blasting the surface to create the bond interface, metal spraying the surface, and machining the inside diameter of the bore.

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3.0 Implementation Steps

3.1. Rough cut: Upper- and lower-cylinder block liner bores must be machined to remove the pitting. Most of the time, cutting off 0.040" to the diameter of the bore must be enough. It depends on how deep pitting is.

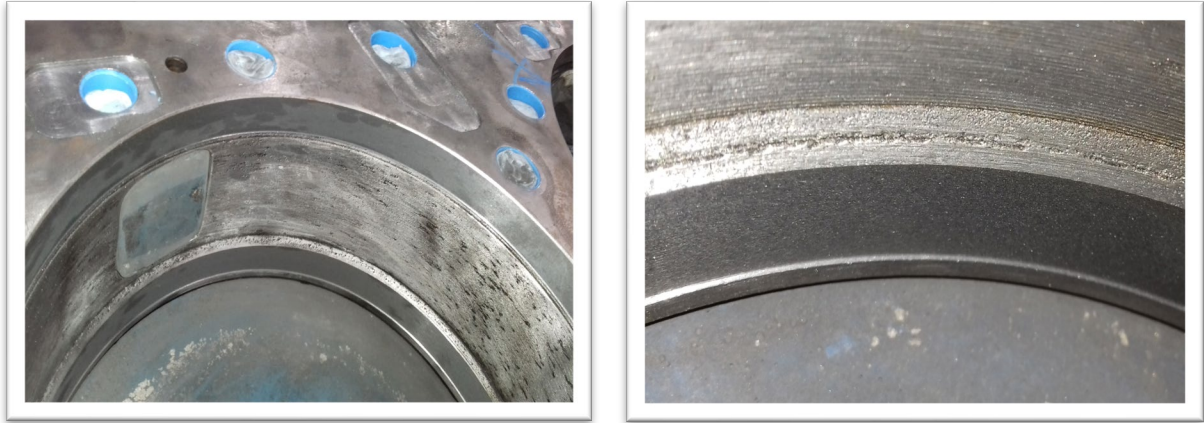


Figure 2. Rough cut on upper and lower seal surfaces

3.2. Top deck bore and thread protection: Use anti-bond materials to cover and protect all the threads and bores which are near to the area to be sprayed.



Figure 3. Top deck bore and thread protected

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3.3. Grit blasting: Follow the instructions in SEBF2120 “Thermal Spray Procedures for Engine Cylinder Block Top Surface”. Propel a stream of aluminum oxide against the block surface under high pressure to prepare the work piece for metal spray.

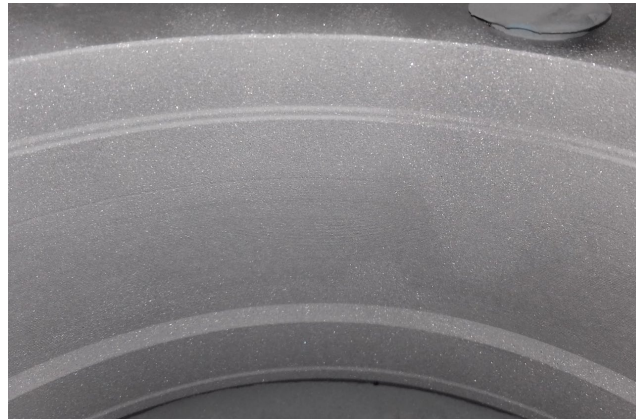


Figure 4. Grit blasted surface

3.4. Metal spraying the surface: Using an industrial robot and metal spray equipment it is possible to control velocity, distance and position of the spray with a high degree of accuracy improving the strength of the bond and finish of the coating. For the spraying parameters use the specifications found in SEBF2120 “Thermal Spray Procedures for Engine Cylinder Block Top Surface”.



Figure 5. Spraying upper- and lower-cylinder block liner bores

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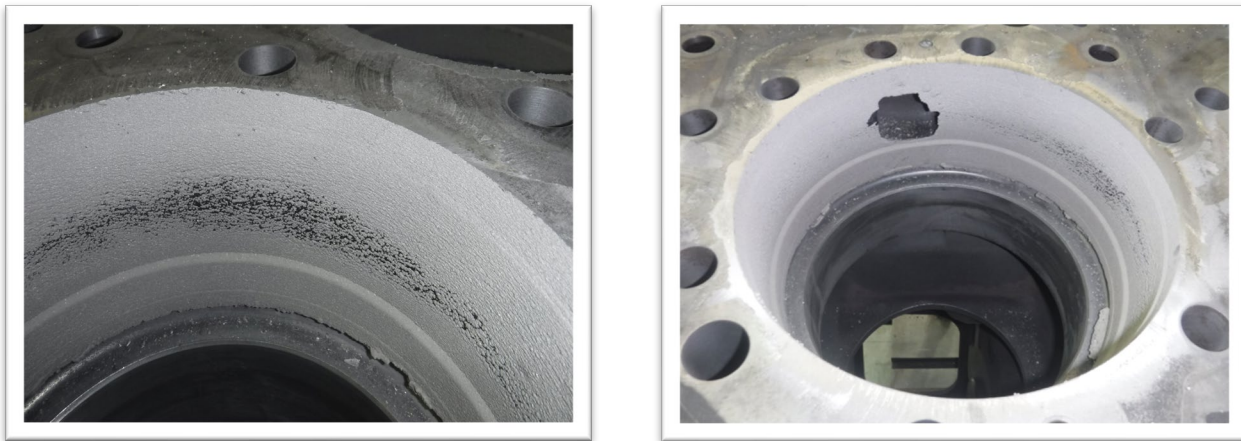


Figure 6. Sprayed upper- and lower-cylinder block liner bores

3.5. Finishing cut: Machine the upper and lower bore surfaces according to the specifications found in the Reuse and Salvage Guidelines SEBF9185 “Specifications and Salvage for Cylinder Blocks in C175 Engines”.

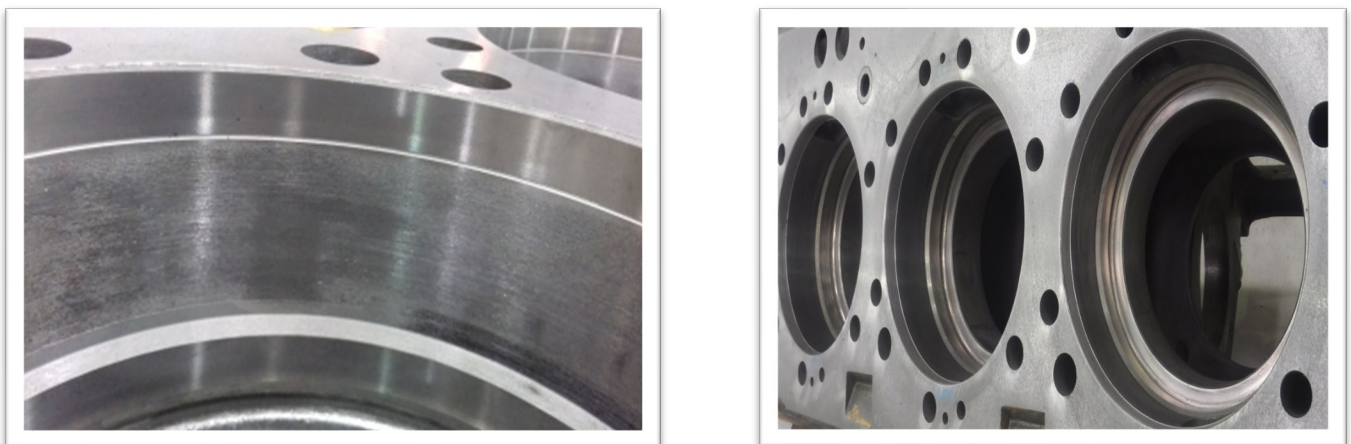


Figure 7. Upper- and lower-cylinder block liner bores repaired

4.0 Benefits

- 4.1. Increases the life of the salvage using wear resistant materials to create the coating.
- 4.2. Eliminates the need to install a machined sleeve when pitting exists on upper- and lower-cylinder block liner bores.
- 4.3. Reduces labor time.

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5.0 Resources Required

- Multi-purpose CNC machining center (Rottler F100 Series or similar)
- Grit blasting equipment
- Industrial robot
- Metal spray equipment

6.0 Supporting Attachments / References

- SEBF9185 “Specifications and Salvage for Cylinder Blocks in C175 Engines”.
- SEBF2120 “Thermal Spray Procedures for Engine Cylinder Block Top Surface”.

7.0 Related Best Practices

- Mounting Fixture for Arc Spray Gun
- Setting-Up the Robot Using a Cone Tool
- Salvage C175 Engine Cylinder Block Upper and Lower Liner Bores Using Metal Spray

8.0 Acknowledgements

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